Yifei Zhang

Applied Scientist, Amazon Alexa AI | Boston, Massachusetts
jimmyzyf@amazon.com | yifei.zhang2@emory.edu
Homepage | Google Scholar | GitHub

Last updated: November 13, 2026

Research Interests

Artificial intelligence and machine learning, with emphasis on evaluation and benchmarking of large language models and agentic AI systems, explainable AI, multimodal learning and reasoning, and privacy-aware AI.

Education

Ph.D. in Computer Science, Emory University Aug 2022 – Mar 2026
Atlanta, Georgia
Degree conferred: May 11, 2026
Advisor: Prof. Liang Zhao
Dissertation: *Benchmarking, Correcting, and Accelerating AI Reasoning across Diverse Modalities*

M.S. in Data Science, Columbia University Sep 2020 – Dec 2021
New York, New York

Master of Engineering Management, Duke University Aug 2019 – May 2020
Durham, North Carolina

B.E. in Engineering Mechanics, Dalian University of Technology Sep 2015 – Jun 2019
Dalian, China

Research and Industry Experience

Amazon Alexa AI Apr 2026 – Present
Applied Scientist *Boston, Massachusetts*

- Conducting research and development on scalable evaluation methods for agentic conversational systems.
- Designing frameworks to assess reliability, task completion, conversational quality, and multimodal behavior.
- Combining diverse evaluation signals into interpretable quality assessments for model and system improvement.
- Researching privacy-aware model adaptation and distillation under practical system constraints.

Amazon Alexa AI May 2025 – Aug 2025
Applied Scientist Intern *Boston, Massachusetts*

- Developed a memory system and multi-agent evaluation framework for Alexa and related agentic conversational AI scenarios.
- Built synthetic dialogue-generation and benchmarking pipelines for long-horizon, multi-turn, and concurrent user tasks.
- Developed evaluation infrastructure for agentic task-oriented dialogue and memory-intensive conversational scenarios.

Amazon AGI Jun 2024 – Sep 2024
Research Scientist Intern *Boston, Massachusetts*

- Developed reward modeling and preference-based evaluation workflows for LLM-driven smart-speaker interactions.
- Built synthetic preference-data generation and parameter-efficient fine-tuning pipelines to support scalable benchmarking and model development.

Emory University Aug 2022 – Mar 2026
Ph.D. Student Researcher *Atlanta, Georgia*

- Conducted doctoral research on evaluation benchmarks and explanation-guided methods for reliable artificial intelligence across language, vision, and multimodal data.

PUBLICATIONS

Full publication list: Google Scholar.

Peer-Reviewed and Accepted Scholarly Works

M. Zhu, S. Cheng, G. Bai, **Yifei Zhang**, and L. Zhao. *Dual Decomposition for Text-to-Image Retrieval and Generation*. Findings of the Association for Computational Linguistics: EACL 2027, accepted. Earlier preprint: *Cross-modal RAG*.

G. Bai, Z. Chai, C. Ling, S. Wang, J. Lu, N. Zhang, T. Shi, Z. Yu, M. Zhu, **Yifei Zhang**, X. Song, C. Yang, Y. Cheng, and L. Zhao. *Beyond Efficiency: A Systematic Survey of Resource-Efficient Large Language Models*. ACM Computing Surveys, vol. 58, no. 16, Article 422, pp. 1–45, 2026. doi:10.1145/3845797.

Yifei Zhang, H. Nayyeri, R. Khaziev, E. Yilmaz, G. Tur, D. Hakkani-Tür, and H. P. Thadakamalla. *ATOD: An Evaluation Framework and Benchmark for Agentic Task-Oriented Dialogue Systems*. AACL-IJCNLP 2026, Main Conference, pp. 2399–2416.

X. Liu, H. Nayyeri, D. Hakkani-Tür, E. Yilmaz, J.-K. Kim, **Yifei Zhang**, C. Peris, and H. Thadakamalla. *Hear2Act: Benchmarking When Prosody Should Change What an Assistant Does*. NeurIPS 2026 RTCA Workshop, accepted (poster).

Yifei Zhang, J. Song, S. Gu, T. Jiang, B. Pan, G. Bai, and L. Zhao. *Saliency-Bench: A Comprehensive Benchmark for Evaluating Visual Explanations*. Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2025.

B. Pan, Z. Xiong, G. Wu, Z. Zhang, Y. Hu, **Yifei Zhang**, and L. Zhao. *GraphNarrator: Generating Textual Explanations for Graph Neural Networks*. Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, 2025.

B. Pan, Z. Zhang, **Yifei Zhang**, Y. Hu, and L. Zhao. *Distilling Large Language Models for Text-Attributed Graph Learning*. Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, 2024.

X. Li, **Yifei Zhang**, and L. Zhao. *Multi-Prompt Fine-Tuning of Foundation Models for Enhanced Biomedical Image Segmentation*. In *AI for Health Equity and Fairness*, Studies in Computational Intelligence, vol. 1164, pp. 83–91. Springer, 2024.

Q. Zhao, **Yifei Zhang**, M. Zhu, S. Gu, Y. Gao, X. Yang, and L. Zhao. *DUE: Dynamic Uncertainty-Aware Explanation Supervision via 3D Imputation*. Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2024.

Yifei Zhang, B. Pan, C. Ling, Y. Hu, and L. Zhao. *ELAD: Explanation-Guided Large Language Models Active Distillation*. Findings of the Association for Computational Linguistics: ACL 2024, 2024.

Yifei Zhang, B. Pan, S. Gu, G. Bai, M. Qiu, X. Yang, and L. Zhao. *Visual Attention Prompted Prediction and Learning*. Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, 2024.

D. Yu, B. Chen, Y. Li, S. Dhakal, **Yifei Zhang**, Z. Liu, M. Zhang, J. Zhang, and L. Zhao. *STES: A Spatiotemporal Explanation Supervision Framework*. Proceedings of the 2024 SIAM International Conference on Data Mining, 2024.

Yifei Zhang, S. Gu, Y. Gao, B. Pan, X. Yang, and L. Zhao. *MAGI: Multi-Annotated Explanation-Guided Learning*. Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023.

S. Gu, **Yifei Zhang**, Y. Gao, X. Yang, and L. Zhao. *ESSA: Explanation Iterative Supervision via Saliency-Guided Data Augmentation*. Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2023.

Preprints and Other Scholarly Manuscripts

Yifei Zhang, B. Pan, M. Zhu, J. Pei, and L. Zhao. *Agentic Commerce: A Survey of How AI Agents Are Reshaping Commerce*. TechRxiv preprint, 2026.

Yifei Zhang, M. Li, H. Gao, and L. Zhao. *EmoLLM: Appraisal-Grounded Cognitive-Emotional Co-Reasoning in Large Language Models*. arXiv preprint arXiv:2603.16553, 2026.

Yifei Zhang, J. Zhang, M. Safari, X. Yang, and L. Zhao. *Explainable Cross-Disease Reasoning for Cardiovascular Risk Assessment from Low-Dose Computed Tomography*. arXiv preprint arXiv:2511.06625, 2025.

Yifei Zhang, T. Jiang, B. Pan, J. Wang, G. Bai, and L. Zhao. *MEGL: Multimodal Explanation-Guided Learning*. arXiv preprint arXiv:2411.13053, 2024.

G. Bai, Q. Zhao, X. Jiang, **Yifei Zhang**, and L. Zhao. *Saliency-Guided Hidden Associative Replay for Continual Learning*. arXiv preprint arXiv:2310.04334, 2023.

B. Pan, Z. Liu, **Yifei Zhang**, and L. Zhao. *SurroCBM: Concept Bottleneck Surrogate Models for Generative Post-hoc Explanation*. arXiv preprint arXiv:2310.07698, 2023.

B. Pan, M. Qin, S. Wang, **Yifei Zhang**, and L. Zhao. *Controllable Data Generation Via Iterative Data-Property Mutual Mappings*. arXiv preprint arXiv:2310.07683, 2023.

Academic Service

Journal Reviewer: ACM TIST; IEEE TBD; IEEE TNNLS; JACMP; Expert Systems with Applications; Information Processing and Management; Neurocomputing; Information Systems Journal.

Conference Reviewer / Program Committee: ICLR 2025–2027; NeurIPS 2026; AAAI 2027; CVPR 2025–2026; ICCV 2025; ECCV 2026; BMVC 2026; ACL ARR 2025–2026; KDD Research Track 2025–2026; KDD Datasets and Benchmarks Track 2025–2026; MemAgents Workshop (ICLR 2026).

Workshop Organization: Publicity Chair, DeepSpatial 2024 (KDD); Student Co-organizer, FM4sBio 2025 (AAAI).

Honors, Grants, and Memberships

- Sigma Xi, Full Membership.
- Tinker Research Grant.
- KDD 2023 Student Travel Award.
- Dalian University of Technology Overseas Alumni Association Seagle Scholarship Award.
- Academic Excellence Scholarship Award, Dalian University of Technology.
- Technological Innovation Scholarship Award, Dalian University of Technology.

Technical Skills

Programming: Python, C/C++, MATLAB, Java, Shell
Systems: Linux, Git, AWS, Slurm, Docker
Frameworks: PyTorch, TensorFlow, Transformers, Accelerate, DeepSpeed, TRL, PEFT